$$
\begin{aligned}
&C_{dG\phi_2}{}^{i1\_i2\_} \to \\
&\frac{1}{16\pi^2}\Big(-\frac{1}{18} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{d}}{}^{i2}] + \frac{1}{9} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{d}}{}^{i2}] - \frac{1}{18} g_3 g_1^2 y_d{}^{i1i2} \\
&\quad \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{d}}{}^{i2}] - \frac{1}{72} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,0}[m_1, m_{\tilde{q}}{}^{i1}] + \frac{1}{36} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,-1}[m_1, m_{\tilde{q}}{}^{i1}] - \\
&\quad \frac{1}{72} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{4,1,-2}[m_1, m_{\tilde{q}}{}^{i1}] - \frac{3}{8} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,0}[m_2, m_{\tilde{q}}{}^{i1}] + \\
&\quad \frac{3}{4} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,-1}[m_2, m_{\tilde{q}}{}^{i1}] - \frac{3}{8} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{4,1,-2}[m_2, m_{\tilde{q}}{}^{i1}] + \\
&\quad \frac{1}{12} g_3^3 y_d{}^{i1i2} \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{d}}{}^{i2}] + \frac{7}{12} g_3^3 y_d{}^{i1i2} \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{d}}{}^{i2}] - \\
&\quad \frac{2}{3} g_3^3 y_d{}^{i1i2} \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{d}}{}^{i2}] + \frac{1}{12} g_3^3 y_d{}^{i1i2} \mathrm{LF}_{2,1,0}[m_3, m_{\tilde{q}}{}^{i1}] + \\
&\quad \frac{7}{12} g_3^3 y_d{}^{i1i2} \mathrm{LF}_{3,1,-1}[m_3, m_{\tilde{q}}{}^{i1}] - \frac{2}{3} g_3^3 y_d{}^{i1i2} \mathrm{LF}_{4,1,-2}[m_3, m_{\tilde{q}}{}^{i1}] - \\
&\quad \frac{1}{4} g_3 \overline{y_d}{}^{pr} y_d{}^{pi2} y_d{}^{i1r} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{d}}{}^{r}] + \frac{1}{2} g_3 \overline{y_d}{}^{pr} y_d{}^{pi2} y_d{}^{i1r} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{d}}{}^{r}] - \\
&\quad \frac{1}{4} g_3 \overline{y_d}{}^{pr} y_d{}^{pi2} y_d{}^{i1r} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{d}}{}^{r}] - \frac{1}{2} g_3 \overline{y_d}{}^{pr} y_d{}^{pi2} y_d{}^{i1r} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{q}}{}^{p}] + \\
&\quad g_3 \overline{y_d}{}^{pr} y_d{}^{pi2} y_d{}^{i1r} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{q}}{}^{p}] - \frac{1}{2} g_3 \overline{y_d}{}^{pr} y_d{}^{pi2} y_d{}^{i1r} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{q}}{}^{p}] - \\
&\quad \frac{1}{4} g_3 y_d{}^{pi2} \overline{y_u}{}^{pr} y_u{}^{i1r} \mathrm{LF}_{2,1,0}[\tilde{\mu}, m_{\tilde{u}}{}^{r}] + \frac{1}{2} g_3 y_d{}^{pi2} \overline{y_u}{}^{pr} y_u{}^{i1r} \mathrm{LF}_{3,1,-1}[\tilde{\mu}, m_{\tilde{u}}{}^{r}] - \\
&\quad \frac{1}{4} g_3 y_d{}^{pi2} \overline{y_u}{}^{pr} y_u{}^{i1r} \mathrm{LF}_{4,1,-2}[\tilde{\mu}, m_{\tilde{u}}{}^{r}] - \frac{1}{18} g_3 m_1 g_1^2 a_d{}^{i1i2} \mathrm{LF}_{2,1,1,0}[m_1, m_{\tilde{d}}{}^{i2}, m_{\tilde{q}}{}^{i1}] + \\
&\quad \frac{1}{18} g_3 m_1 g_1^2 a_d{}^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_1, m_{\tilde{d}}{}^{i2}, m_{\tilde{q}}{}^{i1}] - \frac{1}{6} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{d}}{}^{i2}, \tilde{\mu}] + \\
&\quad \frac{1}{3} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{d}}{}^{i2}, \tilde{\mu}] - \frac{1}{6} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{d}}{}^{i2}, \tilde{\mu}] - \\
&\quad \frac{1}{12} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{1,1,1,0}[m_1, m_{\tilde{q}}{}^{i1}, \tilde{\mu}] + \frac{1}{6} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_1, m_{\tilde{q}}{}^{i1}, \tilde{\mu}] - \\
&\quad \frac{1}{12} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_1, m_{\tilde{q}}{}^{i1}, \tilde{\mu}] - \frac{1}{6} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{d}}{}^{i2}] - \\
&\quad \frac{1}{12} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_1, \tilde{\mu}, m_{\tilde{q}}{}^{i1}] - \frac{3}{4} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{1,1,1,0}[m_2, m_{\tilde{q}}{}^{i1}, \tilde{\mu}] + \\
&\quad \frac{3}{2} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,1,-1}[m_2, m_{\tilde{q}}{}^{i1}, \tilde{\mu}] - \frac{3}{4} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,1,-2}[m_2, m_{\tilde{q}}{}^{i1}, \tilde{\mu}] - \\
&\quad \frac{3}{4} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{2,2,1,-2}[m_2, \tilde{\mu}, m_{\tilde{q}}{}^{i1}] - \frac{1}{6} m_3 g_3^3 a_d{}^{i1i2} \mathrm{LF}_{2,1,1,0}[m_3, m_{\tilde{d}}{}^{i2}, m_{\tilde{q}}{}^{i1}] - \\
&\quad \frac{4}{3} m_3 g_3^3 a_d{}^{i1i2} \mathrm{LF}_{3,1,1,-1}[m_3, m_{\tilde{d}}{}^{i2}, m_{\tilde{q}}{}^{i1}] + \frac{1}{3} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{d}}{}^{i2}] - \\
&\quad \frac{1}{6} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{d}}{}^{i2}] + \frac{1}{6} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_1, m_{\tilde{q}}{}^{i1}] - \\
&\quad \frac{1}{12} g_3 g_1^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_1, m_{\tilde{q}}{}^{i1}] + \frac{3}{2} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{2,1,1,-1}[\tilde{\mu}, m_2, m_{\tilde{q}}{}^{i1}] - \\
&\quad \frac{3}{4} g_3 g_2^2 y_d{}^{i1i2} \mathrm{LF}_{3,1,1,-2}[\tilde{\mu}, m_2, m_{\tilde{q}}{}^{i1}] + \frac{1}{2} g_3 \tilde{\mu}^2 y_d{}^{pi2} \overline{y_u}{}^{pr} y_u{}^{i1r} \mathrm{LF}_{2,1,1,0}[\tilde{\mu}, m_{\tilde{q}}{}^{p}, m_{\tilde{u}}{}^{r}] - \\
&\quad \frac{1}{2} g_3 \tilde{\mu}^2 y_d{}^{pi2} \overline{y_u}{}^{pr} y_u{}^{i1r} \mathrm{LF}_{3,1,1,-1}[\tilde{\mu}, m_{\tilde{q}}{}^{p}, m_{\tilde{u}}{}^{r}]\Big)
\end{aligned}
$$